



IBM Quantum

Multi-Qubit Circuits & Entanglement

Fatima Bint-e-Munir



Qiskit Fall Fest 2026 at University of Engineering & Technology Lahore, Narowal Campus

Speaker Introduction

ABOUT:

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Interested in everything Physics
Nonlinear Phenomenon in Fusion devices, Space
Plasmas and NLTs
[Quantum Hamiltonian Simulations](#)

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FALL FEST ROADMAP

You Are Here: Talk 3 of the Series

✓ **Intro to QC + Qiskit setup**
Why quantum, environment ready

✓ **Vectors, matrices & statevectors**
The math language of qubits

✓ **Single-qubit gates + superposition**
X, Y, Z, H, S, T and the Bloch sphere

🔔 **Multi-qubit systems + entanglement**
This talk – tensor products, Bell & GHZ states
WE ARE HERE

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60-SECOND RECAP

What You Already Know

- **A qubit is a vector**
 $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$
- **Gates are matrices**
Unitary matrices transform one statevector into another
- **The Hadamard gate**
 $H|0\rangle = |+\rangle$ – an equal mix of $|0\rangle$ and $|1\rangle$
- **The Born rule**
 $P(\text{outcome}) = |\text{amplitude}|^2$
How amplitudes become probabilities

From last time

```
from qiskit.quantum_info import Statevector
psi = Statevector.from_label("++") # H|0>
print(psi) # Statevector([0.70710678, 0.70710678j], dims=(2,1))
```

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TODAY'S SESSION

Where We're Headed

- 01 Multi-qubit state representation From one qubit to n qubits – 2^n amplitudes
- 02 Tensor products Mathematical tool for combining quantum systems
- 03 Entanglement + hands-on lab Correlations with no classical explanation
- 04 Break 15 minutes
- 04 Bell states The four canonical entangled pairs
- 05 GHZ states Entanglement across three or more qubits

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BY THE END OF THIS TALK

Learning Objectives

- 01 Write the statevector for a system of n qubits, and read off amplitudes correctly
- 02 Compute the tensor product of two quantum states or operators by hand and in Qiskit
- 03 Explain what entanglement is – and why most multi-qubit states cannot be factored
- 04 Build and verify a Bell state, and describe what makes each of the four Bell states distinct
- 05 Extend the Bell-state circuit to build a GHZ state on three or more qubits

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PART 1

Multi-Qubit State Representation

How do we describe two, three, or a hundred qubits at once?

Billie Marlow, Qubit Fall Term 2020 at University of Engineering & Technology, London, Newark Campus

BUILDING UP GRADUALLY

From One Qubit to Two

A real quantum computer needs many qubits working together. Let's take the smallest possible step up — from one qubit to two — and see how their possibilities combine.

Recap from Task 2: one qubit has exactly two basis states — $|0\rangle$ and $|1\rangle$.

TWO COINS

Flip two coins, A and B. Each one lands heads (H) or tails (T).

	B=H	B=T
A=H	HH	HT
A=T	TH	TT

TWO QUBITS

Each qubit, labeled q_1 and q_2 , is alone either $|0\rangle$ or $|1\rangle$.

	$q_2= 0\rangle$	$q_2= 1\rangle$
$q_1= 0\rangle$	$ 00\rangle$	$ 01\rangle$
$q_1= 1\rangle$	$ 10\rangle$	$ 11\rangle$

Same pattern, new symbols — two independent binary choices always combine into four joint possibilities.

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FROM BITS TO BITSTRINGS

Two Qubits, One Joint System

Writing two bits together as one string — the 01 — makes a bitstring. Qubits combine their labels the exact same way.

CLASSICAL

2 bits → 4 possible bitstrings

$(0,1) + (0,1) = \{00, 01, 10, 11\}$

Just like our two coins: at any moment the pair has exactly ONE definite bitstring value — say 01 — and nothing more.

QUANTUM

2 qubits → 4 basis states

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$

Unlike the coins, a qubit pair can be spread across all four bitstring outcomes at once — with a weight (amplitude) on each.

$(\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle)$

This is the first genuinely new thing multi-qubit systems bring: superposition over JOINT outcomes, not just single-qubit ones. We'll write this out precisely next.

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PART 2

Tensor Products

The one operation that combines separate quantum systems into one

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THE CORE IDEA

What is a Tensor Product?

The tensor product $\otimes$ combines a state of one system and a state of another into one joint state.

$$|\phi\rangle \otimes |\psi\rangle = |\phi\rangle|\psi\rangle = |\phi\psi\rangle$$

three equivalent ways to write the same joint state — the $\otimes$ symbol is often dropped entirely

A product state means independence

When a pair of qubits is in the state $|\phi\rangle \otimes |\psi\rangle$, the first qubit is simply in state $|\phi\rangle$, the second is simply in state $|\psi\rangle$, and the two have nothing to do with one another. Such is called a product state.

Tensor products also generalize to three or more systems: $|\phi_1\rangle \otimes \dots \otimes |\phi_n\rangle$ is a valid joint state whenever each factor is.

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WORKED EXAMPLE

Tensor Products, Step by Step

Basis states: $|0\rangle \otimes |0\rangle$

$$|0\rangle \otimes |1\rangle = |0\rangle_1 |1\rangle_2 = |01\rangle$$

Combining two basis states just concatenates their labels — no arithmetic needed.

Superposition: $\frac{1}{\sqrt{2}}|0\rangle$

$$= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |0\rangle$$

$$= \frac{1}{2} \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

Distribute the tensor product over the sum, just like multiplying out brackets in algebra.

Two amplitudes multiply together

$$|\phi\rangle = a|0\rangle + b|1\rangle, |\psi\rangle = c|0\rangle + d|1\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Every amplitude of the joint state is a product of the amplitudes of original factors

Order matters: $|\phi\rangle \otimes |\psi\rangle$ is generally not equal to $|\psi\rangle \otimes |\phi\rangle$ — tensor products are not commutative.

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THE BUILDING BLOCK

The Two-Qubit State Vector

A general state of two qubits is a normalized column vector with one amplitude per basis state.

$$|\psi\rangle = \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix} = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Normalization still applies

$|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$ — the total probability across all four outcomes still sums to 1.

Example: $\frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$ is a valid two-qubit state.

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A CONVENTION WORTH GETTING RIGHT

Reading Order: Qiskit's Little-Endian

Qiskit numbers qubits starting from 0, and reads them right to left — the opposite of how we read English.

$|10\rangle$ $q_1 = 1$ $q_0 = 0$ Affront digit = highest indexed qubit
rightmost digit = qubit 0

Why it trips people up

If you write `qc.h0()` then `qc.x(1)` on two qubits starting at `|00\rangle`, the resulting state is `|11\rangle` — but Qiskit prints it as `|10\rangle`, with qubit 1 (the X) on the LEFT and qubit 0 (the H) on the right.

Always check which qubit index a digit belongs to before reading a multi-qubit ket.

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SCALING UP

n Qubits, 2^n Amplitudes

Every additional qubit doubles the number of basis states — and the number of amplitudes needed to describe the system.

Qubits	Basis states (2^n)	Amplitudes needed
1	2	α_0, α_1
2	4	$\alpha_{00}, \alpha_{01}, \alpha_{10}, \alpha_{11}$
3	8	$\alpha_{000} \dots \alpha_{111}$
10	1,024	still simulable on a laptop
50	$\approx 1.1 \times 10^{15}$	exceeds every classical supercomputer's memory

This exponential wall is exactly why quantum computers can, in principle, tackle problems no classical machine can simulate directly.

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IN QISKIT

Building & Reading Multi-Qubit States

Qubit — a two qubit statevector

```
from qiskit.quantum_info import Statevector
psi = Statevector([0.5, +0.5j, 0.5j, 0.5])
print(psi) # [0.5+0.j, +0.5+0.j, 0.+0.5j, 0.5+0.j], dims=(2, 2)
```

That statevector prints with dims=(2, 2). Which entry is the amplitude for `|10\rangle` — the 2nd or the 3rd (index 2)?

Remember the ordering: index *k* in binary tells you each qubit's value, qubit 0 on the right.

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IN QISKIT

Tensor Products in Qiskit

Qiskit — Statevector and Operator both have .tensor()

```
from qiskit.quantum_info import Statevector, Operator
psi0 = Statevector.from_label("0")
zero = Statevector.from_label("0")
psi = psi0.tensor(zero)
# Statevector([0.707+0.j, 0.+0.j, 0.707+0.j, 0.+0.j]), dims=(2, 2)
```

The tensor factor on the right

`plus.tensor(zero)` places zero as the right-hand factor — matching `(+) ⊗ |0\rangle`.

The cnot operator (`plus * zero`) is exact shorthand for the same call.

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PREDICT BEFORE YOU PEEK

Try It — Expand a Tensor Product

$|1\rangle \otimes |-\rangle = |1\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - \frac{1}{\sqrt{2}}|1\rangle) = ?$

Expand this product the same way we expanded `|+\rangle ⊗ |0\rangle` earlier. Which two basis states appear, and with what amplitudes?

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► CAN WE ALWAYS GO BACKWARD

Not Every State Is a Product State

We've been building joint states FROM two separate qubits. Does every joint state factor apart like this?

$$1/\sqrt{2} |00\rangle + 1/\sqrt{2} |11\rangle = |\phi\rangle \otimes |\psi\rangle ?$$

Try to find single-qubit states $|\phi\rangle$ and $|\psi\rangle$ that multiply out to this. What goes wrong?

Details in next lecture. But we can't really separate this state into two product states.

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► COMBINING GATES

Tensor Products of Operators

We combined qubit GATES by combining their matrices through tensor product. We combine qubit GATES the same way — by tensoring their full matrices together.

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{a 4x4 matrix}$$

The general rule

If A is an $m \times n$ matrix and B is an $p \times q$ matrix, then $A \otimes B$ is one big $(m \cdot p) \times (n \cdot q)$ matrix — built by replacing every single entry of A with a scaled copy of the ENTIRE matrix B .

$H \otimes I$ means: take every entry of H , and put a full (scaled) copy of I in its place.

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► SHOWING THE WORK

Expanding $H \otimes I$, Block by Block

Replace each entry of H with that entry times the whole matrix I — then just write out what each block looks like.

each block is a full copy of I , scaled by the matching H entry

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \end{bmatrix}$$

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

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► IN QISKIT

Multi-qubit Gates

Qiskit automatically performs the block-expansion we just did by hand.

Qiskit

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(4)
qc.h(0)          qc.i(1) in parallel
qc.draw(output='mpl')
```

In practice, you'll never write this by hand

When you call `qc.h(0)` on a 2-qubit circuit, Qiskit builds $H \otimes I$ automatically under the hood — you just name the qubit, and the tensoring with identity on every other qubit happens for you.

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► GENERALIZING

The Same Pattern, More Qubits

Nothing about tensoring gates is special to two qubits — a single-qubit gate on a bigger register just tensors with identity everywhere else.

$H \otimes I \otimes I \otimes I$

Qiskit

```
qc = QuantumCircuit(3)
qc.h(0)
```

Qiskit builds $H \otimes I \otimes I$ automatically — you only ever name the qubit.

Why it matters

For n qubits, even a single-qubit gate becomes a $2^n \times 2^n$ matrix. That's the same exponential wall from Part 1's scaling table — it's exactly why classical simulation of quantum circuits gets expensive so fast.

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► A NEW KIND OF GATE

Meet CNOT: A Two-Qubit Gate

Every gate so far has acted on one qubit at a time. CNOT is different — it reads one qubit to decide what to do to the other.

Circuit symbol

Truth table

$ 00\rangle$	$\rightarrow$	$ 00\rangle$	control 0, no flip
$ 01\rangle$	$\rightarrow$	$ 11\rangle$	control 0, target flips
$ 10\rangle$	$\rightarrow$	$ 10\rangle$	control 1, no flip
$ 11\rangle$	$\rightarrow$	$ 01\rangle$	control 1, target flips

The rule: Flip the target qubit, but only when the control qubit is 1. If the control is 0, nothing happens at all.

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► GENUINELY NEW

CNOT's Matrix — and Why It's Different

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Unlike every gate so far, this matrix cannot be written as $A \otimes B$ for any single-qubit matrices A and B .

Quick proof sketch
A tensor product $A \otimes B$ always sends a product state to another product state. But C_X sends $|+\rangle \otimes |0\rangle \rightarrow$ a product state $\rightarrow 1/\sqrt{2}(|00\rangle + 1/\sqrt{2}|11\rangle)$, which is entangled.
So C_X cannot be a tensor product of two single-qubit gates.

This is exactly why pairing H with $CNOT$ creates entanglement in Part 3 — a gate that can't be factored is precisely what's needed to produce a state that can't be factored.

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► KEEP THIS HANDY

Tensor Product Cheat Sheet

- Combine states $|\phi\rangle \otimes |\psi\rangle$ — joint state of two independent systems
- Combine operators $U_1 \otimes U_2$ — independent gates acting together
- Multiply amplitudes Each joint amplitude = one factor from each state, multiplied multiplied
- Order matters $\otimes$ is not commutative — keep qubit order consistent

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► PART 3

Entanglement

Correlations between qubits that no classical explanation can account for

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► SETTING EXPECTATIONS

A Correlation, Not a Mystery

Once called "spooky action at a distance", is now being tested in labs each day.

TWO MATCHED GLOVES

Send two gloves in boxes, guaranteed to always match, match, and ship them to different cities. Opening one one instantly tells you the other. Each glove was a part of part of the pair the whole time; you just didn't know the know the left or right.

ENTANGLEMENT — WITH ONE REAL DIFFERENCE

Entangled qubits show the same instant correlation. The genuine surprise, however, is that qubits do NOT have a hidden predetermined value like the gloves. The correlation is real but nothing waiting to be revealed.

This is a scientific fact, not folklore: entanglement is confirmed by decades of Bell test experiments — work recognized by the 2022 Nobel Prize in Physics.

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► THE DEFINITION

What Is Entanglement?

A joint state is entangled precisely when it is NOT a product state.

SEPARABLE (product state)

$|\phi\rangle \otimes |\psi\rangle$

Qubit 1 is definitely in some state $|\phi\rangle$, qubit 2 is definitely in some state $|\psi\rangle$. They can be described completely on their own, independent of one another.

ENTANGLED (not separable)

$1/\sqrt{2} |00\rangle + 1/\sqrt{2} |11\rangle$

Neither qubit has its own definite state. The two outcomes are perfectly correlated — measuring one instantly tells you the other — with no way to describe either qubit alone.

For quantum state vectors, entanglement and correlation are the same thing: not a product state $\Rightarrow$ entangled.

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► CLOSING THE LOOP FROM PART 2

Proving It Can't Be Split

Suppose, for contradiction, that $1/\sqrt{2}(|00\rangle + 1/\sqrt{2}|11\rangle) = |\phi\rangle \otimes |\psi\rangle$ for some single-qubit states.

- Look at the $|00\rangle$ amplitude** $(|0\rangle\langle 0| \otimes |1\rangle\langle 1|)$ must equal 0, since $|01\rangle$ doesn't appear. So $\langle 0|\phi\rangle = \langle 0|\psi\rangle = 0$.
- But look at $|00\rangle$ and $|11\rangle$** $(|0\rangle\langle 0| \otimes |0\rangle\langle 0|) = 1/2$ and $(|1\rangle\langle 1| \otimes |1\rangle\langle 1|) = 1/2$ — both nonzero, so neither $\langle 0|\phi\rangle$ nor $\langle 1|\psi\rangle$ can be zero.
- Contradiction** No such $|\phi\rangle$ and $|\psi\rangle$ can exist. The state is genuinely entangled — this isn't a limit of our search, it's a proof.

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► FROM THEORY TO CIRCUIT

Creating Entanglement: H • CNOT

Two gates are all it takes: a Hadamard to create superposition, then a controlled NOT to correlate it with a second qubit.



Why the CNOT entangles

CNOT flips the target only when the control is 1. Because the control is in superposition, the flip and the no-flip both happen at once, in step with the control — that's exactly what tangles the two qubits together.

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► THE BIGGER PICTURE

Why Entanglement Matters

- It broke classical intuition: The Bell state's correlations are strong enough to rule out any "hidden variable" explanation — entangled qubits are not just hiding a pre-agreed answer.
- It scales — genuinely: IBM researchers have verified genuine multipartite and bipartite entanglement across many qubits of a processor — real hardware, not just theory.
- It's a computational resource: Teleportation, superdense coding, and most quantum algorithms that outperform classical ones all consume entanglement as fuel.

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► TIME FOR A BREAK

Back in 15 Minutes

When we return: Bell states, then GHZ states — entanglement across two across two qubits, then three or more.



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► HANDS-ON LAB - 15 MINUTES

Let's Experience It Hands-On

Time to build and measure a real entangled pair yourselves



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► PART 4

Bell States

The four canonical entangled pairs, named for physicist John Stewart Bell



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► THE BELL BASIS

The Four Bell States

Every one of these is maximally entangled — and together, they form a complete basis for two qubits.

$ \Phi^+\rangle$	$1/\sqrt{2} (00\rangle + 1/\sqrt{2} 11\rangle)$	$ \Phi^-\rangle$	$1/\sqrt{2} (00\rangle - 1/\sqrt{2} 11\rangle)$
$ \Psi^+\rangle$	$1/\sqrt{2} (01\rangle + 1/\sqrt{2} 10\rangle)$	$ \Psi^-\rangle$	$1/\sqrt{2} (01\rangle - 1/\sqrt{2} 10\rangle)$

This collection is called the Bell basis — any two-qubit state can be written as a combination of these four.

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► THE RECIPE

Building $|\Phi^+\rangle$, Step by Step

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
 $-CX \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |\Phi^0\rangle$

- 1. Start in $|00\rangle$**
Both qubits initialize to $|0\rangle$ by default in Qiskit.
- 2. Hadamard on q_0**
Puts the control qubit into $|+\rangle$, an equal superposition.
- 3. CNOT, $q_0 \rightarrow q_1$**
Ties q_1 to whatever q_0 becomes — entanglement complete.

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► IN QISKIT

A Bell State in Four Lines

```

from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)

print(Statevector(qc))
# Statevector([0.70710678+0.j, 0.70710678+0.j, 0.70710678+0.j, 0.70710678+0.j], dims=[2, 2]) - this is |Phi>

```

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► FROM THEORY TO DATA

Measuring a Bell State

1,000 shots of $|\Phi^+\rangle$, measured and tallied — only two outcomes ever appear.

Outcome	Count
00	497
01	0
10	0
11	503

Perfectly correlated
 $|01\rangle$ and $|10\rangle$ never appear at all. Every single shot lands on 00 or 11 — roughly 50/50, but always matching.
 That's the signature of entanglement: measuring one qubit tells you the other's outcome with certainty, even though each individual outcome is random.
 Run it yourself — the exact split will differ each time.

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► PREDICT BEFORE YOU PEEK

Try It — From $|\Phi^+\rangle$ to $|\Psi^+\rangle$

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$? $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

What single one-qubit gate, applied to q_0 right after the CNOT, turns $|\Phi^+\rangle$ into $|\Psi^+\rangle$?
Think in terms of bits: $2 \otimes 2 = 4$ terms. Flip only the q_0 half of each term. It on q_0 turns $|00\rangle \rightarrow |01\rangle$ and $|11\rangle \rightarrow |10\rangle$ — exactly $|\Psi^+\rangle$.

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► THE BIGGER PICTURE

Why Bell States Matter

- Bell tests**
Measuring Bell states in different bases produces correlations too strong for any string for any classical, local-hidden-variables theory to explain.
- Quantum teleportation**
A shared Bell pair lets you transmit an unknown qubit's state using only classical communication plus entanglement.
- Superdense coding**
A single qubit, if it's half of a Bell pair, can carry two classical bits of information instead of one.

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► PART 5

Teleportation

Correlations between qubits that no classical explanation can account for

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BEFORE WE GO FURTHER

Disclaimers

Teleportation is real and useful — but it isn't what science-fiction promises. Four myths to clear up before we dive in.

- No — we can't teleport a person to another galaxy.**
- It's not about something disappearing and reappearing elsewhere.**
- It doesn't work instantaneously — no information travels faster than light.**
- It doesn't retain information or matter at both places at once.**

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A FUNDAMENTAL LIMIT

No-Cloning Theorem

"It is impossible to create an identical copy of an arbitrary unknown quantum state."

you cannot duplicate an unmeasured qubit the way you can copy a classical bit.

- Copying requires reading**
You can't copy a qubit without first reading — measuring — what's being copied.
- Reading destroys the state**
Measurement collapses this superposition, leaving only the classical outcome — the original quantum state is gone.
- Teleportation moves, not clones**
Quantum teleportation transfers an unknown state to a new qubit — but destroys the original qubit in the process.

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PUTTING BELL PAIRS TO WORK

Quantum Teleportation: The Protocol

Alice wants to send Bob her qubit's exact state $|\psi\rangle$ — without moving the qubit, and without ever copying it.

- Alice and Bob share a Bell pair in advance — one qubit each.
- Alice entangles her qubit with her half, then measures both — yielding bits m_0, m_1 .
- Alice sends those 2 bits to Bob over an ordinary classical channel.
- Bob applies the matching correction — his qubit becomes exactly $|\psi\rangle$.

Bob's correction:

m_0, m_1	Gate
00	I
01	X
10	Z
11	XZ

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CLOSING THE LOOP

Why This Isn't Magic

- Nothing outruns light**
Only 2 classical bits ever travel from Alice to Bob. Until they arrive, Bob's qubit is just noise — exactly what the Disclaimers promised.
- Nothing is cloned**
The instant Alice measures, her original qubit collapses — its state is gone from her side from that side forever. There is only ever ONE copy, exactly as the Cloning requires.
- Nothing disappears and reappears**
Bob's physical qubit was there the whole time; only its STATE changes, via two ordinary two ordinary corrective gates.

Every disclaimer turns out to be exactly true — teleportation never breaks a rule we've observed.

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PART 5

GHZ States

Entanglement stretched across three qubits or more, named for Greenberger, Horne & Zeilinger

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GENERALIZING ENTANGLEMENT

From Two Qubits to Three

Nothing about entanglement stops at two qubits — the same idea extends to any number.

$$|GHZ\rangle = \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |111\rangle$$

All three qubits, perfectly tied together

Just like the Bell state, the GHZ state is not a product state — you cannot describe any one qubit on its own. Measure any single qubit and you get 0 or 1 with 50/50 odds, but the moment you do, the other two collapse to match it exactly.

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THE RECIPE, EXTENDED

Building a GHZ State

Same pattern as the Bell state — one Hadamard, then a CNOT fanning out from that same control qubit to every other to every other qubit.

q₀: does all the work

The Hadamard gate q_0 , in preparation, each CNOT then copies that superposition onto one more qubit, using q_0 as the central both times. Add a third Cx for a fourth qubit, and so on — the pattern scales to any n .

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COMPARE & CONTRAST

GHZ vs. Bell

BELL STATE (2 qubits)
 $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

- One CNOT, one pair entangled
- 4 basis states, 2 appear
- Losing one qubit destroys the pair entirely

GHZ STATE (3 qubits)
 $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$

- $n-1$ CNOTs, all fanned from one control
- 2 ^{n} basis states, only 2 ever appear
- Losing just one qubit still leaves the rest correlated as a smaller GHZ-like state

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FROM THEORY TO DATA

Measuring a GHZ State

```

from qiskit import QuantumCircuit

qc = QuantumCircuit(3)
qc.h(0); qc.cx(0, 1); qc.cx(0, 2)
qc.measure_all()
# 1000 shots -- "half "000", "half "111"

```

Only two of eight possible outcomes ever occur

Out of 8 possible 3-bit outcomes, only 000 and 111 ever appear — never a mix like 010 or 101. All three qubits agree, every single time, exactly like the Bell state but now with a third entry added to the side.

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PREDICT BEFORE YOU PEEK

Try It — Measuring Just One Qubit

$|GHZ\rangle = \frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle$ — measure only q_0 — what happens to q_1 and q_2 ?

? If measuring q_0 gives 0, what state are q_1 and q_2 left in? What if it gives 1?

Each outcome has probability 1/2. Getting 0 collapses everything to $|000\rangle$; getting 1 collapses everything to $|111\rangle$ — the other two qubits were never independent to begin with.

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RECAP

Let's Check the List

- Write the statevector for a system of n qubits, and read off amplitudes correctly
- Compute the tensor product of two quantum states or operators by hand and in Qiskit
- Explain what entanglement is — and why most multi-qubit states cannot be factored
- Build and verify a Bell state, and describe what makes each of the four Bell states distinct
- Extend the Bell-state circuit to build a GHZ state on three or more qubits

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KEEP EXPLORING

Resources

- IBM Quantum Learning — Multiqubit Systems**
The most advanced today's vector products, entanglement, Bell & GHZ states quantum.cloud.ibm.com/learning
- Qiskit API documentation**
The full reference for every gate, primitive, and visualization function quantum.cloud.ibm.com/docs
- Qiskit source & examples**
Browse real circuits and contribute to the open-source project github.com/Qiskit/qiskit
- IBM Quantum Composer**
Drag-and-drop circuit building in your browser, no install needed quantum.cloud.ibm.com/composer

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▶ THANK YOU

Questions?

No pressure, no wrong questions — if entanglement still feels strange, that means you're paying attention.



Dr. A. K. Singh, Global Fall Fest 2026, at University of Engineering & Technology, Lucknow, Personal Campus

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